

```
#sentimen Analysis English
```

```
clothing <- read.csv("Review Dress.csv")
```

```
dim(clothing)
```

```
library(dplyr)
```

```
glimpse(clothing)
```

```
clothing_1 <- clothing %>% select(Rating, Class.Name, Review.Text)
```

```
head(clothing_1)
```

```
# Jumlah data unik untuk Rating dan Class.Name
```

```
nrow(unique(as.data.frame(clothing_1$Rating)))
```

```
# Jumlah data unik untuk Rating dan Class.Name
```

```
nrow(unique(as.data.frame(clothing_1$Rating)))
```

```
#Jumlah data kosong
```

```
sum(is.na(clothing_1))
```

```
#Data yang akan digunakan dalam analisis project ini adalah data teks yang berada pada  
rating rendah,
```

```
#yakni: nilai rating 1 dan 2 dan
```

```
#untuk kategori product dengan comment terbanyak (asumsi product utama) yakni  
Dresses.
```

```
clothing_rating_low <-
```

```
clothing_1 %>%
```

```
filter(Rating==1|Rating==2) %>%
```

```
filter(Class.Name=="Dresses")
```

```
#Jumlah data observasi
```

```
nrow(clothing_rating_low)
```

```
# Take text data only
```

```
clothing_rating_low_text <- clothing_rating_low[,3]
```

```
install.packages("tm")
```

```
#library corpus da
```

```
library(tm)
```

```
# Create a corpus
```

```
clothing_rating_low_text_corpus <- Corpus(VectorSource(clothing_rating_low_text))
```

```
head(clothing_rating_low_text_corpus$content, 3)
```

```
# Preprocessing Data
```

```
clothing_rating_low_text_corpus <-
```

```
  clothing_rating_low_text_corpus %>%
```

```
  tm_map(content_transformer(tolower)) %>% # Case-folding
```

```
  tm_map(content_transformer(removePunctuation)) %>% # Remove punctuation
```

```
  tm_map(content_transformer(removeNumbers)) %>% # Remove numbers
```

```
  tm_map(content_transformer(lemmatize_strings)) %>% # Stemming & lemmatize
```

```
  tm_map(removeWords, stopwords("english")) # Remove stopwords
```

```
# Mengubah data ke bentuk dgCMatrix class
```

```

clothing_rating_low_text_corpus_dtm <- clothing_rating_low_text_corpus %>%
DocumentTermMatrix()

inspect(clothing_rating_low_text_corpus_dtm)

#membuat model LDA

rating_low_dtm_lda <- Matrix::Matrix(as.matrix(clothing_rating_low_text_corpus_dtm),
sparse = T)

set.seed(123)

install.packages("textmineR")

#library LDA

library(textmineR)

rating_low_lda <- FitLdaModel(rating_low_dtm_lda,
                             k = 5,
                             iterations = 1000,
                             burnin = 500,
                             calc_coherence = T
)

glimpse(rating_low_lda)

# rating_low_dtm_lda <- Matrix::Matrix(as.matrix(clothing_rating_low_text_corpus_dtm),
sparse = T)

#

# k_list <- seq(10, 30, by = 10)

#

# model_list <- TmParallelApply(X = k_list, FUN = function(k){

```

```

#
# m <- FitLdaModel(dtm = rating_low_dtm_lda,
#                 k = k,
#                 iterations = 10,
#                 burnin = 5,
#                 calc_coherence = TRUE)
#
# m <- mean(m$coherence)
#
# return(m)
# },
# cpus = 4
# )

#Melihat terms yang paling banyak muncul pada setiap topik

# To get the top terms for each topic, we can use the GetTopTerms function.
rating_low_lda_word_topic <- GetTopTerms(rating_low_lda$phi, 50) %>%
  as.data.frame()
# %>% set_names(paste("Topic", 1:5))

rating_low_lda_word_topic

install.packages(c(
  "ggplot2",

```

```

"dplyr",
"tidyr",
"tibble",
"ggwordcloud",
"magrittr",
"rlang"
))
library(ggplot2)
library(dplyr)
library(tidyr)
library(tibble)
library(ggwordcloud)
library(magrittr)

z <-
  rating_low_lda_word_topic %>%
  rownames_to_column("id") %>%
  mutate(id = as.numeric(id)) %>%
  pivot_longer(-id, names_to = "topic", values_to = "term") %>%
  ggplot(aes(label = term, size = rev(id), alpha = rev(id))) +
  geom_text_wordcloud (seed = 123) +
  facet_wrap(~topic, scales = "free") +
  scale_alpha_continuous(range = c(0.4, 1)) +
  theme_minimal() +
  theme(strip.background = element_rect(fill = "blue"),
        strip.text.x = element_text(colour = "white"))

```

z

```
rating_low_lda_word_topic %>%  
  select(t_1) %>%  
  rownames_to_column("id") %>%  
  pivot_longer(-id, names_to = "topic", values_to = "term") %>%  
  ggplot(aes(label = term, size = rev(id), alpha = rev(id))) +  
  geom_text_wordcloud (seed = 123) +  
  theme_minimal()
```